THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 9–13: **Cell: The Building Block of Life, Tissues in Action, Patterns in Life: Diversity and Classification, Reproduction: How Life Continues, Earth as a System: Energy, Matter and Life** | Class 9 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** hypertonic	Water leaves.
2	**(d)** increase in girth	Secondary growth.
3	**(d)** Protista	Euglena.
4	**(a)** vegetative propagation by suckers	Banana is seedless.
5	**(b)** 500 s	d/v.
6	**(a)** 8	Same number.
7	**(b)** squamous epithelium	Alveoli.
8	**(b)** gills throughout life	Fish breathe with gills.
9	**(c)** bisexual flower	Has both.
10	**(a)** coastal low-lying areas	Flooding.
11	**(c)** mitochondria	Energy.
12	**(b)** hinge joint	One plane.
13	**(a)** Animalia, Chordata, Mammalia, Carnivora, Felidae, Panthera, tigris	Hierarchy.
14	**(a)** 46	23 + 23.
15	**(b)** evaporation	Heating.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	RBC has no cell wall (1); Plant cell wall resists the expansion (1).	1+1
17	They are actively dividing (1); Dense cytoplasm and thin walls (1).	1+1
18	Protista (1); Unicellular eukaryote (1).	1+1
19	They come from a single parent (1); Genetically identical clones (1).	1+1
20	t = 500 s (1); = 8.3 minutes (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	In water: mass increases (1); In salt solution: mass decreases (1); Osmosis through membranes (1).	1+1+1
22	Water enters, guard cells swell (1); Stoma opens (1); Water leaves, they become flaccid and the stoma closes (1).	1+1+1
23	Banyan: dicot angiosperm; rice: monocot (1); Fern: pteridophyte; moss: bryophyte (1); Spirogyra: alga; pine: gymnosperm (1).	1+1+1
24	Sperm: small, motile, tail (1); Ovum: large, non-motile, stored food (1); Sperm: DNA delivery; ovum: nutrition for zygote (1).	1+1+1
	OR: Snow reflects 85 per cent (1); Forest and ocean reflect little, absorb a lot (1); Melting of snow increases warming (1).	1+1+1
25	Bacterium has no nucleus, only a nucleoid (1); Animal cell has a nucleus with chromosomes (1); Bacteria have cell walls (1).	1+1+1
	OR: Plants: xylem and phloem (1); Animals: blood and lymph (1); Plants have meristems; animals have dividing cells everywhere (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Double membrane with pores (1); Nucleolus: ribosome synthesis (1); Chromatin: DNA and proteins (1); Control of cell activities (1); Heredity and division (1).	1+1+1+1+1

	OR: Thin section, stain and mount (1); Epidermis: protection (1); Cortex: storage and support (1); Vascular bundles: conduction (1); Pith: storage (1).	1+1+1+1+1
27	Mushroom: Fungi; bacterium: Monera (1); Euglena: Protista (1); Rose: angiosperm; fern: pteridophyte; moss: bryophyte (1); Whale: mammal; snake: reptile; pigeon: bird (1); Earthworm: annelid (1).	1+1+1+1+1
	OR: Gametes formed in testes and ovaries (1); Fertilisation in the oviduct (1); Implantation in the uterus (1); Development with placenta (1); Birth after about 9 months (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 9–13: **Cell: The Building Block of Life, Tissues in Action, Patterns in Life: Diversity and Classification, Reproduction: How Life Continues, Earth as a System: Energy, Matter and Life** | Class 9 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** growing region	Meristem.
2	**(c)** vascular cambium	Rings.
3	**(b)** monocot	Monocot.
4	**(c)** absent	Pregnancy.
5	**(d)** nitrogen fixation	Bacteria and lightning.
6	**(b)** swell and may burst	Hypotonic environment.
7	**(c)** sclerenchyma	Dead and strong.
8	**(d)** fungus	Saprophyte.
9	**(b)** grafting	Joining two varieties.
10	**(a)** using public transport and saving energy	Reduces emissions.
11	**(a)** 4	Half number.
12	**(a)** trachea	Moves mucus.
13	**(a)** bird	Aves.
14	**(d)** unisexual	Separate male and female.
15	**(c)** 700 W per square metre	70 per cent absorbed.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Mitosis: 12 (1); Meiosis: 6 (1).	1+1
17	Phloem is removed (1); Food cannot reach the roots (1).	1+1
18	Breathes by lungs, has mammary glands (1); Warm-blooded and gives birth (1).	1+1
19	Formed by mitosis from the parent body (1); No gametes involved (1).	1+1
20	Rhizobium in root nodules fix nitrogen (1); Soil gets nitrates (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Aves (1); Hollow bones for flight (1); Warm-blooded (1).	1+1+1
22	Hen: eggs with shells, external development (1); Humans: viviparous, development in uterus (1); Both internal fertilisation (1).	1+1+1
23	Overuse of fertilisers (1); Burning fuels produces nitrogen oxides (1); Run-off leads to eutrophication (1).	1+1+1
24	Carry impulses over long distances (1); Branches receive and send signals (1); Structure suits function (1).	1+1+1
	OR: Tissues: epithelial, muscle, connective (1); Organ: stomach (1); System: digestive (1).	1+1+1
25	Flat, bilaterally symmetrical (1); Tapeworm and liver fluke (1); Cause diseases (1).	1+1+1
	OR: Protect mother and baby (1); Balanced diet for growth (1); Regular check-ups (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Energy received = 2 kWh (1); Snow absorbs 0.3 kWh (1); Forest absorbs 1.8 kWh (1); Difference 1.5 kWh (1); Explains why forests are warmer (1).	1+1+1+1+1
	OR: Nucleus: manager (1); Ribosomes: workers (1); ER and Golgi: transport and packing (1); Mitochondria: power (1); Lysosomes: waste disposal; membrane: gate (1).	1+1+1+1+1
27	Aquatic: aerenchyma (1); Desert: thick cuticle, sunken stomata (1); Land: strong xylem (1); Root tissues (1); Examples (1).	1+1+1+1+1

	OR: Birds: feathers, hollow bones, wings (1); Streamlined body (1); Fish: fins, gills (1); Swim bladder, scales (1); Streamlined (1).	1+1+1+1+1

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 9–13: **Cell: The Building Block of Life, Tissues in Action, Patterns in Life: Diversity and Classification, Reproduction: How Life Continues, Earth as a System: Energy, Matter and Life** | Class 9 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** chloroplast	Plants only.
2	**(c)** cartilage and synovial fluid	Lubrication.
3	**(a)** more closely related than species in the same family but different genera	Closer relation.
4	**(c)** 46	Same number.
5	**(d)** 1000 tonnes	20 x 50.
6	**(d)** mitosis	Mitosis.
7	**(b)** secondary xylem	Seasonal growth.
8	**(a)** monocot seed	One cotyledon.
9	**(a)** the 14th day of a 28-day cycle	Mid-cycle.
10	**(b)** run-off	Surface flow.
11	**(a)** protein synthesis	Ribosomes make proteins.
12	**(a)** collenchyma	Corner thickening.
13	**(d)** chordate	Chordata.
14	**(d)** vegetative reproduction	New plantlets.
15	**(b)** converting N2 into nitrogen oxides	High energy.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Rough has ribosomes; smooth lacks (1); Rough makes proteins; smooth lipids (1).	1+1
17	Bone is hard; cartilage is flexible (1); Bone has Ca; cartilage has no blood vessels (1).	1+1
18	Roundworm: cylindrical, unsegmented (1); Earthworm: segmented body (1).	1+1
19	To survive unfavourable conditions (1); Germinate when conditions are right (1).	1+1
20	Ice melts, lowering albedo (1); Warming is amplified (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Persist in the geosphere (1); Pollute the hydrosphere and harm marine life (1); Enter food chains (1).	1+1+1
22	As a cell grows, volume rises faster than surface area (1); Materials cannot reach the interior fast enough (1); Many small cells exchange better (1).	1+1+1
23	Simple: one type of cells, e.g. parenchyma (1); Complex: more than one type, e.g. xylem (1); Complex conduct materials (1).	1+1+1
24	Earthworm: Annelida; hydra: Cnidaria (1); Octopus: Mollusca; tapeworm: Platyhelminthes (1); Starfish: Echinodermata; cockroach: Arthropoda (1).	1+1+1
	OR: Pollen grain germinates on the stigma to form a pollen tube (1); The pollen tube carries the male gamete to the ovule (1); The male gamete fuses with the egg to form a zygote (1).	1+1+1
25	Polar stratospheric clouds form in the cold (1); CFCs release chlorine (1); Chlorine destroys ozone in the sunlight of spring (1).	1+1+1
	OR: Cell size is not related to the body size (1); Larger organisms have more cells (1); Cell number differs (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Seed coat (1); Cotyledons (1); Embryo: radicle, plumule (1); Endosperm in maize (1); Germination steps (1).	1+1+1+1+1

	OR: Forest absorbs 600 tonnes (1); Cars emit 1440 tonnes (1); Forest offsets less than half (1); Need for emission cuts (1); Plant more trees (1).	1+1+1+1+1
27	Diffusion of O2 and CO2 (1); Osmosis of water (1); Active transport against gradient needs energy (1); Example: root cells take up minerals (1); Selective permeability (1).	1+1+1+1+1
	OR: Xylem and phloem together in a bundle (1); Stem: on the same radius (1); Root: on alternate radii (1); Xylem towards the centre (1); Phloem towards the outside (1).	1+1+1+1+1

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